

# Lilly Sweet

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## EDUCATION

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**University of California, Berkeley** MS/PhD in Mechanical Engineering Aug 2025 – Dec 2026  
**GPA:** 3.92 **Graduate Coursework:** Continuum, Rheology, Adv. CFD & Heat Transfer, Adv. Thermo, Adv. Fluids

**University of Kansas** BS in Mechanical Engineering Jan 2021 – May 2025  
**GPA:** 3.98 (*with distinction*) **Involvement:** Aero: Formula SAE

## WORK EXPERIENCE

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**SpaceX** Aero/Thermal Engineer Hawthorne, CA Sep 2026 – Present

- Conducting aerodynamics/thermal analysis for the Falcon Rocket, Dragon Capsule, Starfall, and special projects.

**SpaceX** Falcon Graduate Engineer Hawthorne, CA Jun 2026 – Aug 2026

- Executed **FUN3D CFD** simulations to obtain pressure and heat maps for a quick disconnect aero ramp and V4 Mini tile cover. Ran thermal analysis using CFD heat maps to confirm/size **TPS systems** for both hardware.
- Preprocessed, executed, and post-processed FUN3D CFD simulations simulating the wind tunnel experiments with multiple angles of attack. Found high corroboration between **wind tunnel** and CFD results. Tared out the CFD's sting impacts on aerodynamic coefficients, found agreement between Starfall CFD and flight data.
- Developed a Newtonian theory MATLAB script to analyze stability on simplified Starshield satellite models, enabling aero-stability analysis across multiple configurations to assess atmospheric demise.

**SpaceX** Raptor Graduate Engineer Hawthorne, CA May 2025 – Aug 2025

- Integrated an **AI camera system** to automate machining process inspections, eliminating human-eye subjectivity and achieving **5–20× faster** throughput with improved repeatability and reliability. Utilized RobotStudio.
- Automated** ingestion and organization of **nozzle extension test data** into an MQTT-backed database, enabling engineers to rapidly query results and validate propulsion hardware quality.

**Tesla** Controls Engineering Intern Fremont, CA May 2024 – Aug 2024

- Led a **PLC migration** effort covering >\$200K in hardware and labor, minimizing downtime while modernizing controls infrastructure for long-term reliability.

**Burns & McDonnell** Mechanical Engineering Intern Kansas City, MO May 2023 – Aug 2023

- Developed **automated engineering workflow tools** by combining Python and PowerShell scripting, enabling unattended execution of complex analysis routines previously performed manually or in Excel.

**US Army** Combat Medic Sergeant (M-Day) Jan 2020 – Jan 2026

- Security Clearance:** Secret **Rank:** Sergeant (E-5) (Non-Commissioned Officer) **Honorable Discharge**

## ENGINEERING PROJECTS

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- 2D Viscous (DNS) with Unstructured Meshing FVM Solver:** A custom solver that simulates supersonic flow over a NACA0012. Roe's flux, Superbee & MC limiters (2<sup>nd</sup> order w/ LS), able to successfully resolve shocks.
  - n<sup>th</sup> Order FVM Euler Solver:** Used spectral, FR/CPR, and DG methods to achieve n<sup>th</sup> order accuracy on 1D Euler flow through a converging/diverging nozzle.
  - FSAE Aerodynamics:** Ran iterative STAR-CCM+ simulations. Improved the team's process for higher fidelity simulations. Mechanically designed and manufactured mounting and wings.

## SKILLS

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- Analysis:** STAR-CCM+, Patran/Nastran, Ansys (Mechanical, Fluent, Thermal), Thermal Desktop, Pointwise
  - Coding:** Python, MATLAB, C++/Arduino, Machine Learning, HPC
  - CAD/General:** SolidWorks, NX, ABB Robot Studio, PLC's (ABB, Siemens), Ignition, Carbon Fiber Layups